A-Z Linear Algebra & Calculus for AI, Data Science and Machine Learning

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Regression Analysis in Machine Learning

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Regression analysis in machine learning is a **supervised learning technique** used to model the relationship between a **dependent variable (target)** and one or more **independent variables (features)**. The goal is to **predict continuous values** based on the input data.

Here's the gist:

- **Input:** Historical data with known outputs (e.g., stock price, house prices, temperatures).
- **Output:** A model that can predict future values for unseen inputs.

🔍 Key Concepts:

- **Dependent variable (Y):** What you're trying to predict (e.g., price of a house).
- **Independent variables (X):** The inputs or features (e.g., number of rooms, square footage).
- **Regression model:** A function that maps inputs to predicted outputs.

🛒 Scenario: Predicting House Prices

You work for a real estate company and want to build a machine learning model that predicts the **price of a house** based on its features.

Square Footage	Bedrooms	Bathrooms	Location	Price ($)
2000	3	2	Suburb	300,000
1500	2	1	City	350,000
2500	4	3	Suburb	400,000

- **Independent variables (Features):** Square Footage, Bedrooms, Bathrooms, Location
- **Dependent variable (Target):** Price ($)

📊 Scenario: Predicting Stock Prices for a Company

Let’s say you want to predict the **next day's stock price** for Apple based on historical trends.

Date	Open	High	Low	Close	Volume
2025-01-01	180.00	182.00	178.00	181.00	40,000,000
2025-01-02	181.50	183.00	180.00	182.50	35,000,000
2025-01-03	182.00	185.00	181.00	183.50	45,000,000

Algorithm Type	Description
❑ **Linear Regression**	Assumes a straight-line relationship between input(s) and output.
❑ **Multiple Linear Regression**	Like linear, but with multiple input features.
❑ **Polynomial Regression**	Fits a curved line to the data.
❑ **Ridge/Lasso Regression**	Linear regression with regularization to avoid overfitting.
❑ **Logistic Regression**	Despite the name, it's used for classification (not regression).
❑ **Decision Tree / Random Forest Regression**	Non-linear models based on tree structures.
❑ **Support Vector Regression (SVR)**	Uses SVM principles to predict continuous values.

Linear Regression in Machine Learning

Watch Tutorial

Say you want to predict the **price of a house** based on its **square footage**.

Square Footage (X)	Price ($Y)
1000	200000
1500	250000
2000	300000
2200 (Testing)	**?**

Price = (Slope × Square Footage) + Intercept

Price(y) = mx + c

Mean-Based Formula for OLS:

$$m = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

Price(y) = mx + c

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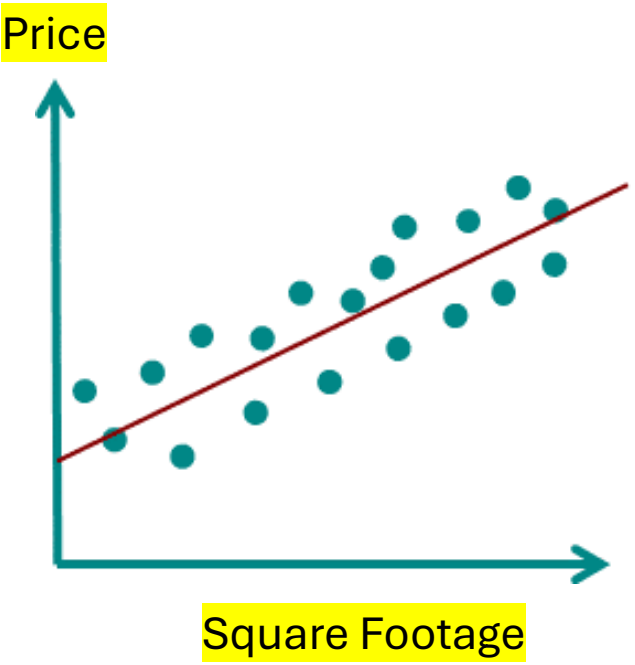

Loss: Original_Price(y) - Price(y)

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Loss: Original_Price(y) - Price(y)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad \text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

How to compute m1, m2, m..., and c in multiple variable linear regression?

⚠️ The Problem with Mean-Based Formulas

Machine Learning

In simple linear regression (1 input), you can use:

$$m = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \quad \text{and} \quad c = \bar{y} - m\bar{x}$$

- r_{12} = correlation between x_1 and x_2
- s_1, s_2 = standard deviations of x_1 and x_2

For two features: $$m_1 = \frac{\sum(x_1 - \bar{x}_1)(y - \bar{y}) - r_{12} \cdot s_1/s_2 \cdot \sum(x_2 - \bar{x}_2)(y - \bar{y})}{\sum(x_1 - \bar{x}_1)^2 \cdot (1 - r_{12}^2)}$$

For **multiple features**, things get messy:

- You'd need to calculate **partial correlations**, **standard deviations**, and **adjust for multicollinearity** (how features influence each other).
- The formulas for m1, m2 become **long and complex**.
- Doesn't scale well to 3, 5, 10, or 100 **features**.
- Easy to make mistakes when features are correlated.

It **does change** and gets **way more complex** due to the interaction between features.

What is the Solution?

Linear Regression

The 'Normal Equation' for Linear Regression

Use matrix form: $\mathbf{w} = (X^T X)^{-1} X^T y$

X is the matrix of inputs: $X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}$

- First column = all 1s (for the intercept c)
- Each row = one data point
- x_{ij} = value of the **j-th feature** for the **i-th example**

The **first column of 1's** is **not the value of c**. It's a **placeholder column** to **learn C** (the **intercept**) during matrix multiplication.

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Produced Result:

$$\mathbf{w} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$y = m_1 x_1 + m_2 x_2 + c$$

The **first column of 1's** is **not the value of c**. It's a **placeholder column** to **learn C** (the **intercept**) during matrix multiplication.

These are the values that go into your final regression equation: $y = m_1 x_1 + m_2 x_2 + c$

$$y = m_1 x_1 + m_2 x_2 + \underbrace{c}_{\text{bias/intercept}}$$

Let’s see an example of math!

Linear Regression

Problem Statement

x_1 (sqft)	x_2 (bedrooms)	y (price)
1000	2	200000
1500	3	250000
2000	4	300000

Let's solve with matrix: $\mathbf{w} = (X^T X)^{-1} X^T y$

$$X = \begin{bmatrix} 1 & 1000 & 2 \\ 1 & 1500 & 3 \\ 1 & 2000 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 200000 \\ 250000 \\ 300000 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1000 & 1500 & 2000 \\ 2 & 3 & 4 \end{bmatrix}$$

Produced Result:

$$\mathbf{w} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$y = m_1 x_1 + m_2 x_2 + c$$

Correlation Analysis

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Correlation Analysis in machine learning (ML) is a statistical method used to measure the strength and direction of the relationship between two or more variables. It's a crucial step in **exploratory data analysis (EDA)** to understand how features in your dataset are related.

🔍 Why is Correlation Analysis Important in ML?

- **Feature Selection**: Helps identify which features are strongly related to the target variable.
- **Multicollinearity Detection**: Helps detect when two or more independent variables are highly correlated, which can affect model performance, especially in linear models.
- **Insights & Understanding**: Gives a clearer picture of relationships in your data.

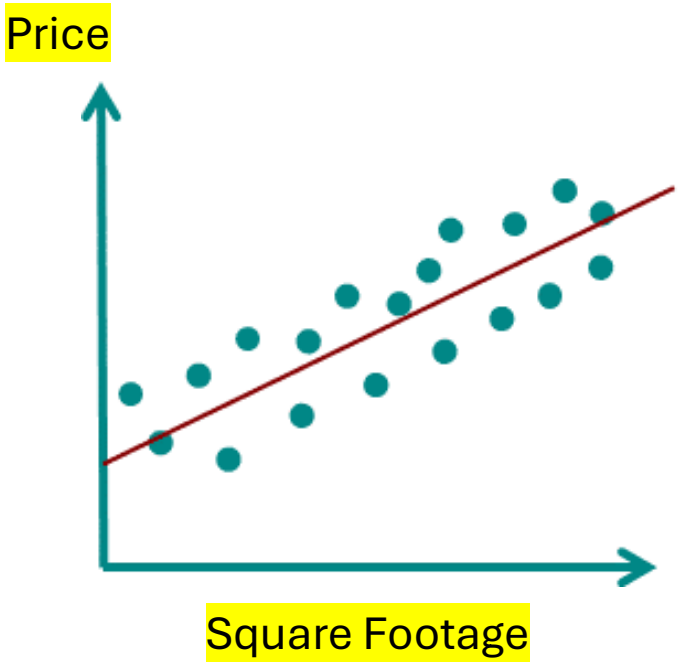
📊 Common Correlation Metrics

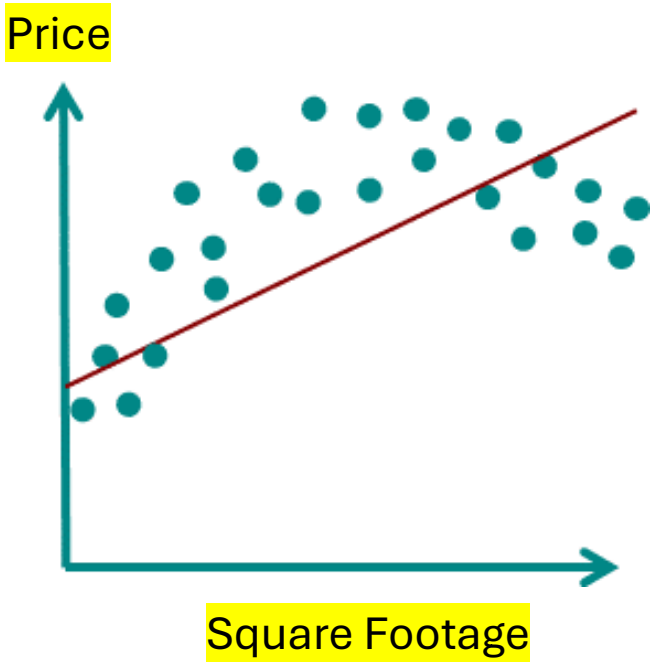
Metric	Best For	Range	Notes
Pearson	Linear relationships	-1 to 1	Most common; assumes normality
Spearman	Monotonic relationships	-1 to 1	Uses rank-order; non-parametric
Kendall Tau	Ordinal data	-1 to 1	More robust for small datasets

Coefficient (r)	Interpretation
1.0	Perfect positive correlation
0.7 to 0.9	Strong positive
0.4 to 0.6	Moderate positive
0.1 to 0.3	Weak positive
0	**No correlation**
-0.1 to -0.3	Weak negative
-0.4 to -0.6	Moderate negative
-0.7 to -0.9	Strong negative
-1.0	Perfect negative correlation

Correlation in Regression

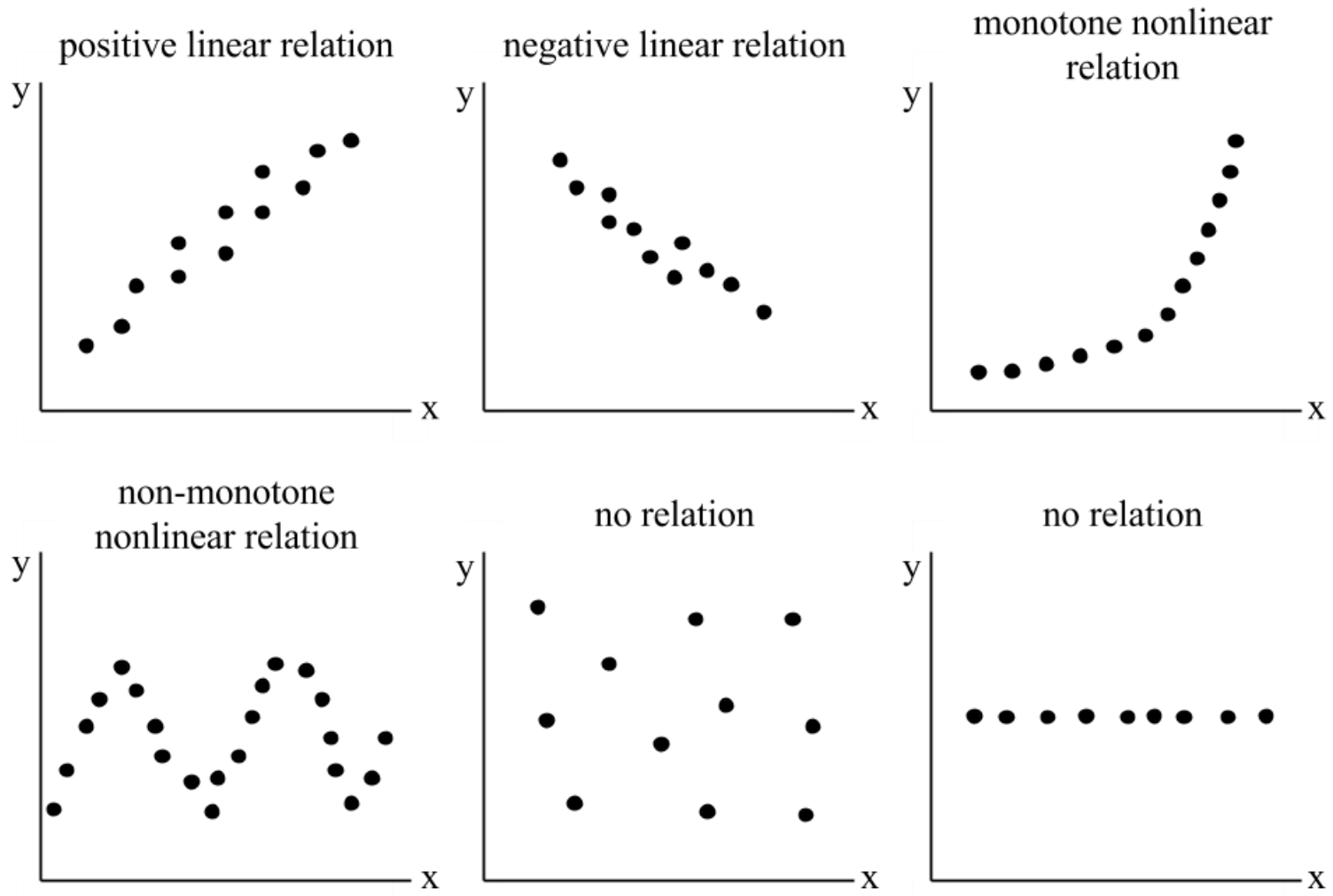
Machine Learning





Correlation in Regression

Machine Learning



Correlation in Regression

Pearson Correlation Coefficient

Pearson Correlation Coefficient: Correlation is always between two variables at a time. When we show a correlation matrix, it just lists all the pairwise correlations between every **pair of variables**.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2} \cdot \sqrt{\sum(y_i - \bar{y})^2}}$$

Where:

- x_i, y_i are individual values of the two variables
- $\bar{x}, \bar{y}$ are their means
- r is the **correlation coefficient** (between -1 and 1)

Dataset

Sample	Size (x)	Price (y)	Age (z)
1	1400	250000	5
2	1600	300000	10
3	1800	350000	15

An example correlation table with **4 variables** commonly found in a housing dataset, showing how they might relate to House Price:

Feature	House Price	Size (sqft)	Number of Rooms	Age of House
House Price	1.00	0.88	0.75	-0.65
Size (sqft)	0.88	1.00	0.70	-0.40
Number of Rooms	0.75	0.70	1.00	-0.30
Age of House	-0.65	-0.40	-0.30	1.00

Interpretation: House Price has-

- a **strong positive** correlation with **Size (sqft)** (0.88)
- a **moderate positive** correlation with **Number of Rooms** (0.75)
- a **strong negative** correlation with **Age of House** (-0.65), meaning older homes tend to be cheaper.

Optimization: Gradient Descent

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Can I Perform Linear Regression using Gradient Descent?

Suppose your linear regression model is: $\hat{y} = w^T x + b$

Your loss function is typically **MSE**: $J(w,b) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$

This cost function is **convex**, so gradient descent will converge to the **global minimum**

1. Compute the gradients:

$$\frac{\partial J}{\partial w} = \frac{2}{n}X^T(Xw + b - y)$$

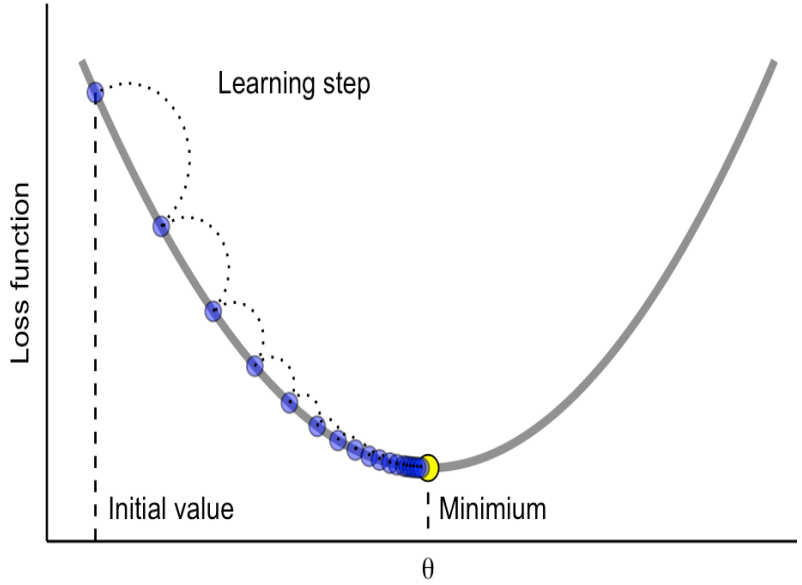
$$\frac{\partial J}{\partial b} = \frac{2}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

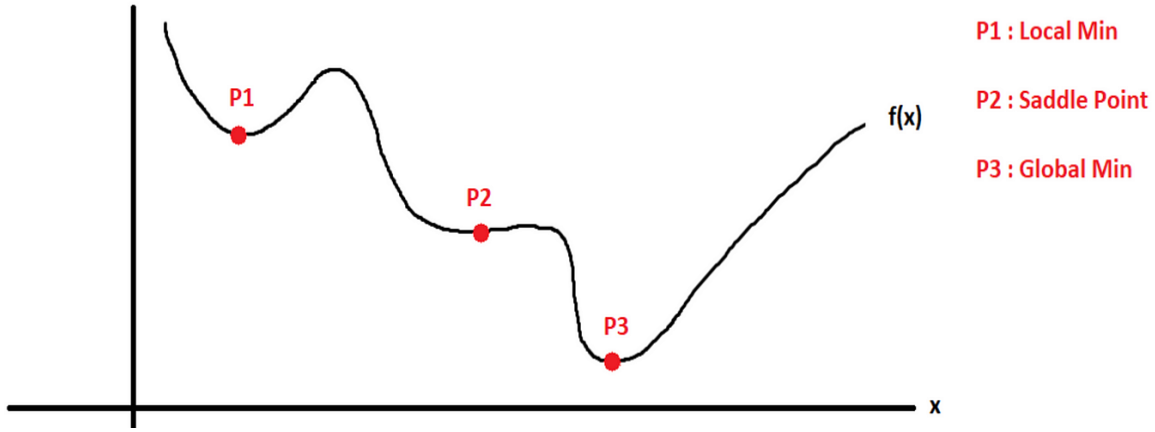
2. Update the parameters: $w := w - \alpha\frac{\partial J}{\partial w}$ $b := b - \alpha\frac{\partial J}{\partial b}$

Gradient Descent in ML & DL

Model Optimization

Gradient descent is a first-order **iterative optimization algorithm** for finding the minimum of a function. Gradient descent can be performed on any differentiable loss function. The main goal of gradient descent is to minimize a cost or loss function, optimizing model parameters for better performance.

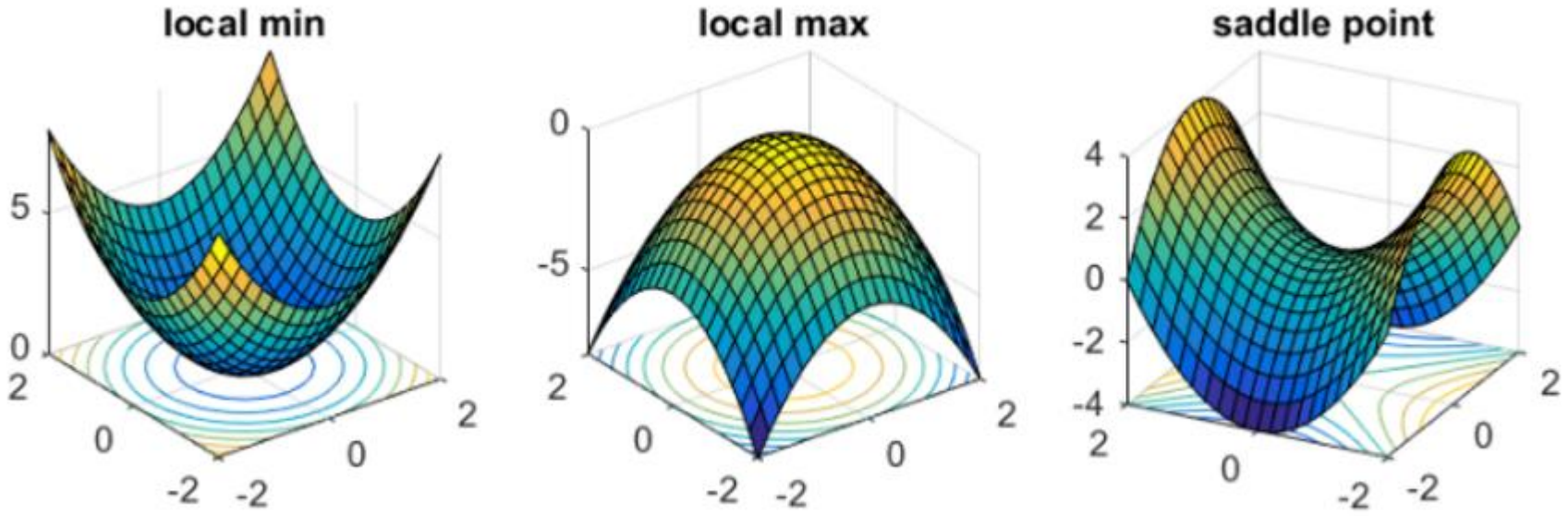




Convexity doesn't depend on MSE alone; It depends on the combination of the cost **function and the model**.

Gradient Descent in ML & DL

Model Optimization



The main equation for updating parameters in gradient descent is:

$$x_{\text{next}} = x - \alpha \cdot \nabla f(x) \quad \textbf{Or,} \quad w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial J}{\partial w}$$

where:

- x_{next} is the updated parameter value,
- x is the current parameter value,
- α is the learning rate,
- $\nabla f(x)$ is the gradient of the function f at x.

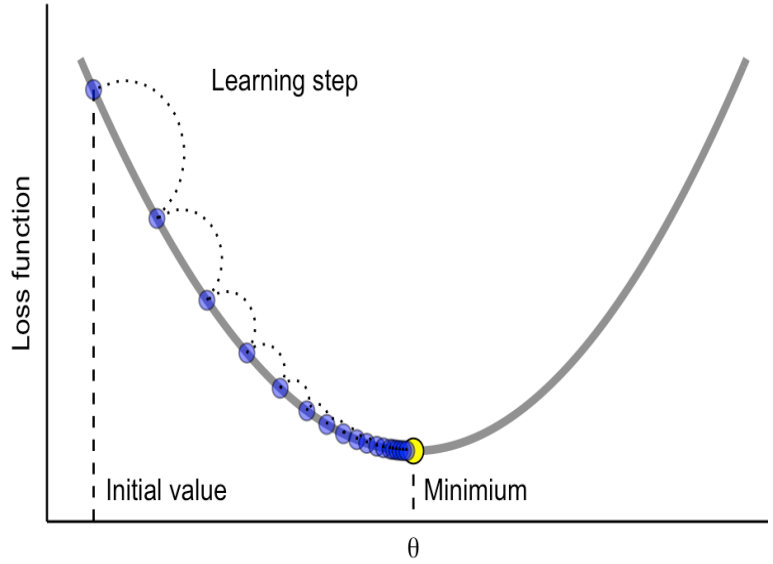


Fig: Gradient Descent

Gradient Descent in ML & DL

Model Optimization

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial J}{\partial w}$$

where:

- w_{new} is the updated weight.
- w_{old} is the current weight.
- α is the learning rate.
- $\frac{\partial J}{\partial w}$ is the gradient of the loss function with respect to the weight.

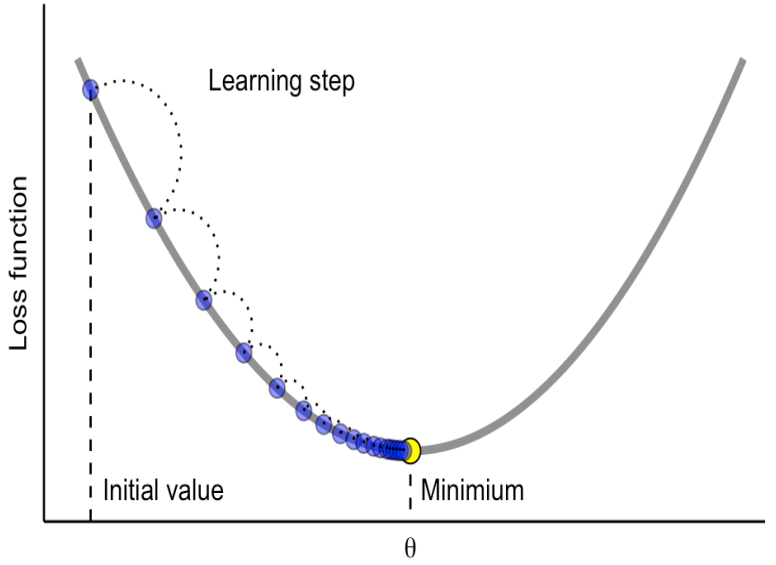


Fig: Gradient Decent

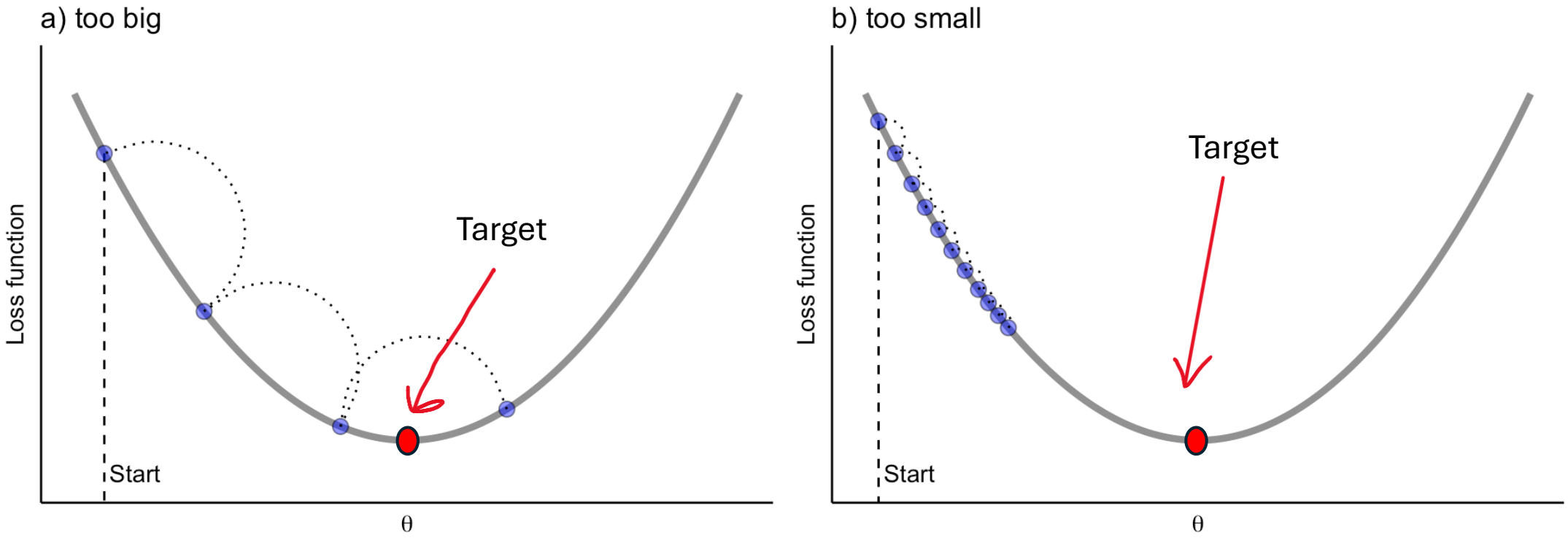
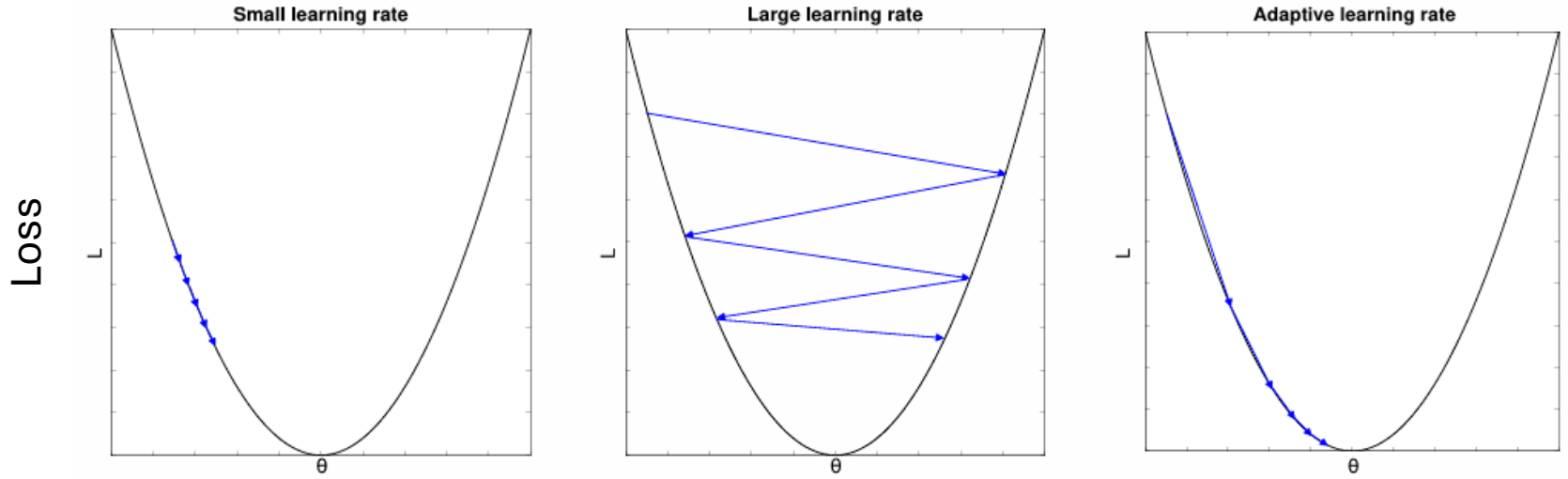


Fig: Importance of Learning Rate

Gradient Descent in ML & DL

Model Optimization



- η too small: long training time
- η too large: miss optima
- Practice: “learning rate decay”: adapt η gradually (e.g.: start with $\eta = 0.01$ and divide every *x* epoch by 10)

Variants of Gradient Descent

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Polynomial Regression

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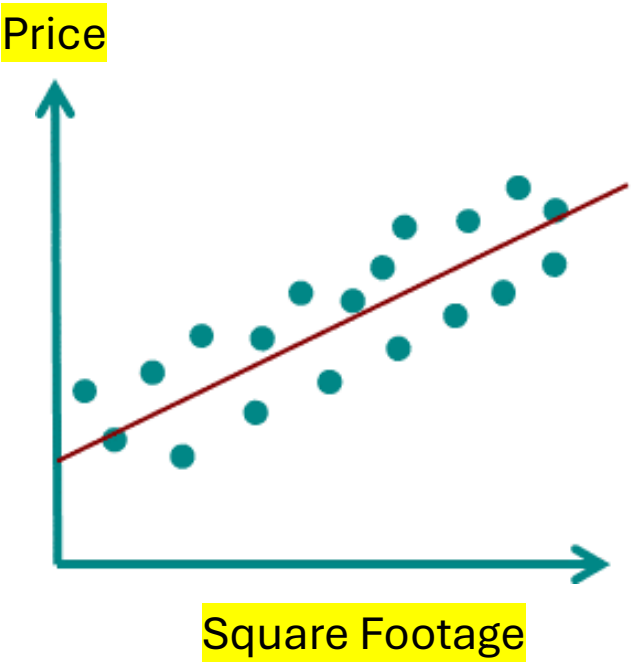
Price(y) = mx + c



Loss: Original_Price(y) - Price(y)

Why Do We Need **Polynomial Regression**?

Because sometimes, real-world data isn't a straight line.

And we need something a ***little curvier*** to capture its true pattern.

Polynomial Regression

Regression Analysis

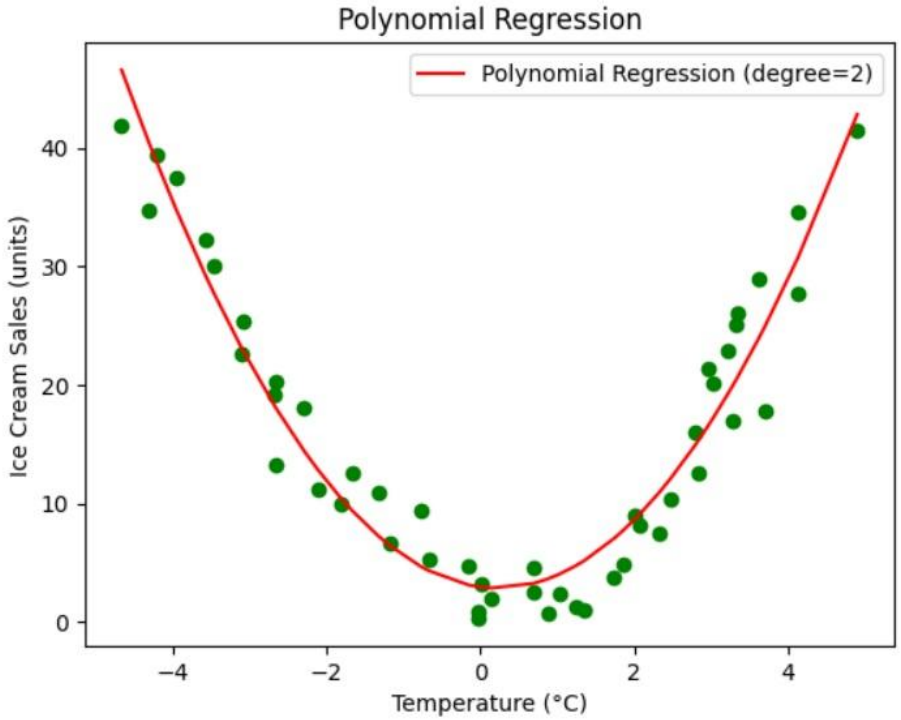
Polynomial regression is a **regression analysis** in which the **relationship between the independent variable x** and the **dependent variable y** is modeled as an **nth-degree polynomial**.

In simple terms:

It's a way to fit a **curved line** to data points, rather than just a straight line (like in linear regression).

When do you use it?

- When the data shows a **non-linear trend**.
- When linear regression doesn't fit the data well.



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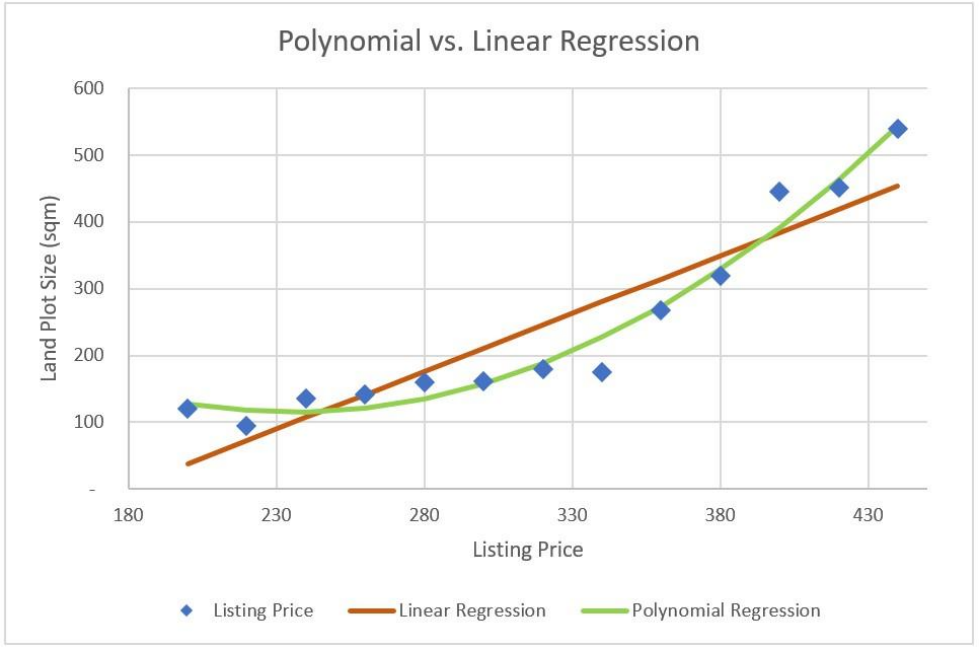

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$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots + \beta_n x^n + \varepsilon$$

Where:

- y: predicted value
- x: input variable
- $\beta_0, \beta_1, \ldots, \beta_n$: coefficients
- ε: error term
- n: the degree of the polynomial

The **degree** is the **highest power of the input variable(s)** in the polynomial function.

- Low degree → High bias, low variance (may miss patterns)
- High degree → Low bias, high variance (may overfit noise)

- 1st Degree: **Linear Regression:** $y = \beta_0 + \beta_1 x$

  Shape: Straight line

- 2nd Degree: **Quadratic Regression:** $y = \beta_0 + \beta_1 x + \beta_2 x^2$

  Shape: Parabola (U-shaped or ∩-shaped)

- 3rd Degree: **Cubic Regression:** $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$

  Shape: Complex curves with multiple bends

$$\text{Number of features} = \binom{n+d}{d} = \frac{(n+d)!}{d! \cdot n!}$$

Say you have n features and degree d; the number of generated features (including interactions and powers) is:

If You Have 1 Feature (univariate):

Degree	Total Features
1	1
2	2 (x, x^2)
3	3 (x, x^2, x^3)
4	4 (x, x^2, x^3, x^4)

$$\binom{1+1}{1} = \binom{2}{1} = 2$$

So, you get **2 terms**:

- One for the bias (constant term)
- One for the feature x

Polynomial Regression

Regression Analysis

$$\text{Number of features} = \binom{n+d}{d} = \frac{(n+d)!}{d! \cdot n!}$$

Degree (d)	Features (n)	Total Terms (with bias)	Generated Terms (examples)
1	1	$\binom{1+1}{1} = 2$	$1, x_1$
2	1	$\binom{1+2}{2} = 3$	$1, x_1, x_1^2$
2	2	$\binom{2+2}{2} = 6$	$1, x_1, x_2, x_1^2, x_1x_2, x_2^2$
3	2	$\binom{2+3}{3} = 10$	$1, x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, x_1x_2^2, x_2^3$
3	3	$\binom{3+3}{3} = 20$	20 total combinations including all interactions and powers

Coefficient of Determination In Regression Analysis

Watch Tutorial

Say you want to predict the **price of a house** based on its **square footage**.

Square Footage (X)	Price ($Y)
1000	200000
1500	250000
2000	300000
2200 (Testing)	**350000**

Price = (Slope × Square Footage) + Intercept

Price(y) = mx + c

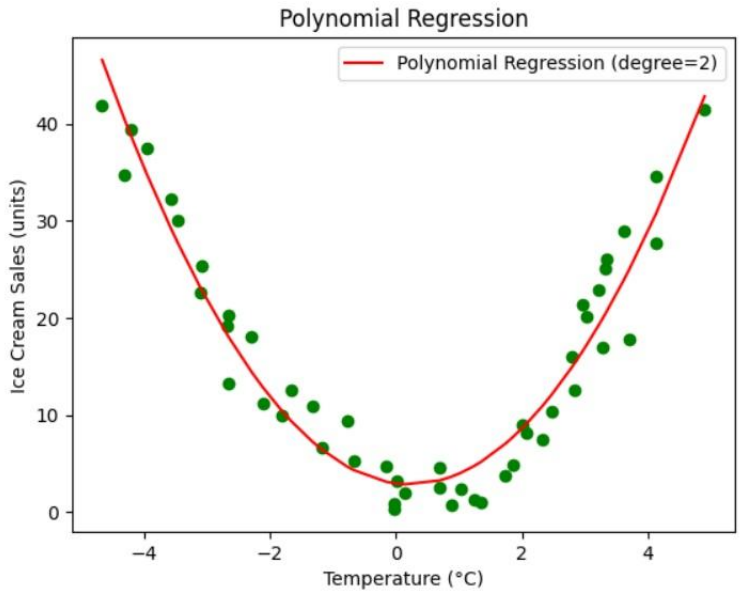
If your goal is to **predict price**, then the main objective is for your predicted values to be **as close as possible** to the **actual/original prices**.

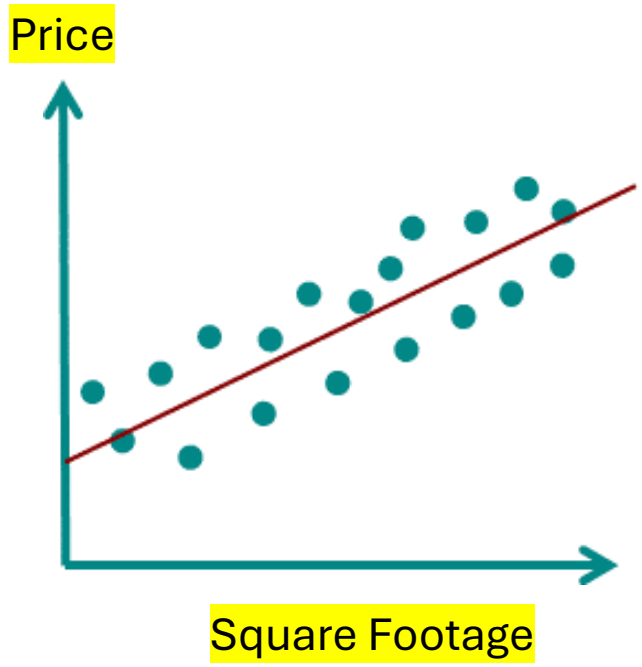
Coefficient of Determination

Regression Analysis

The **coefficient of determination**, often denoted as **R^2 (R-squared)**, is a statistical measure used to assess how well a regression model explains the variability of the dependent variable.

It tells you **how much of the variation in the output (Y)** can be **explained by the input features (X)** in a regression model. R^2 measures how well **any regression model,** including linear, polynomial, and neural network, explains the **variance in the target variable**, with 1 being a perfect fit. Range: $-\infty < R^2 \leq 1$





Coefficient of Determination

Regression Analysis

Formula: $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$

- SS_{res} = Sum of Squares of Residuals
- SS_{tot} = Total Sum of Squares

Interpretation:

- **$R^2 = 1$**: Perfect fit. The model explains 100% of the variance.
- **$R^2 = 0$**: The model explains none of the variance—no better than just using the mean.
- **$R^2 < 0$**: Model is worse than just predicting the mean (can happen with bad models).

Cost Functions

Regression Analysis

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Logistic Regression

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Logistic regression is a statistical method for predicting the **probability of a certain outcome** based on one or more input variables. It's mainly used when the result you're predicting is **binary**, meaning it has two possible values (like yes/no, 0/1, good/bad, success/failure).

Probability= $$\frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}}$$

where:

- e is Euler's number (about 2.718),
- $\beta_0, \beta_1, \ldots$ are the parameters the model learns,
- $x_1, x_2, \ldots$ are the input features.

- Why it’s called Regression?
- Why it's still a Classification method?

Logistic regression is called "regression" because it models a relationship between input variables and a continuous outcome, but the continuous outcome it models is the probability of belonging to a class (between 0 and 1), not the class itself.

Logistic regression does **start** similarly to linear regression: It calculates a **linear combination** of the input features, like:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

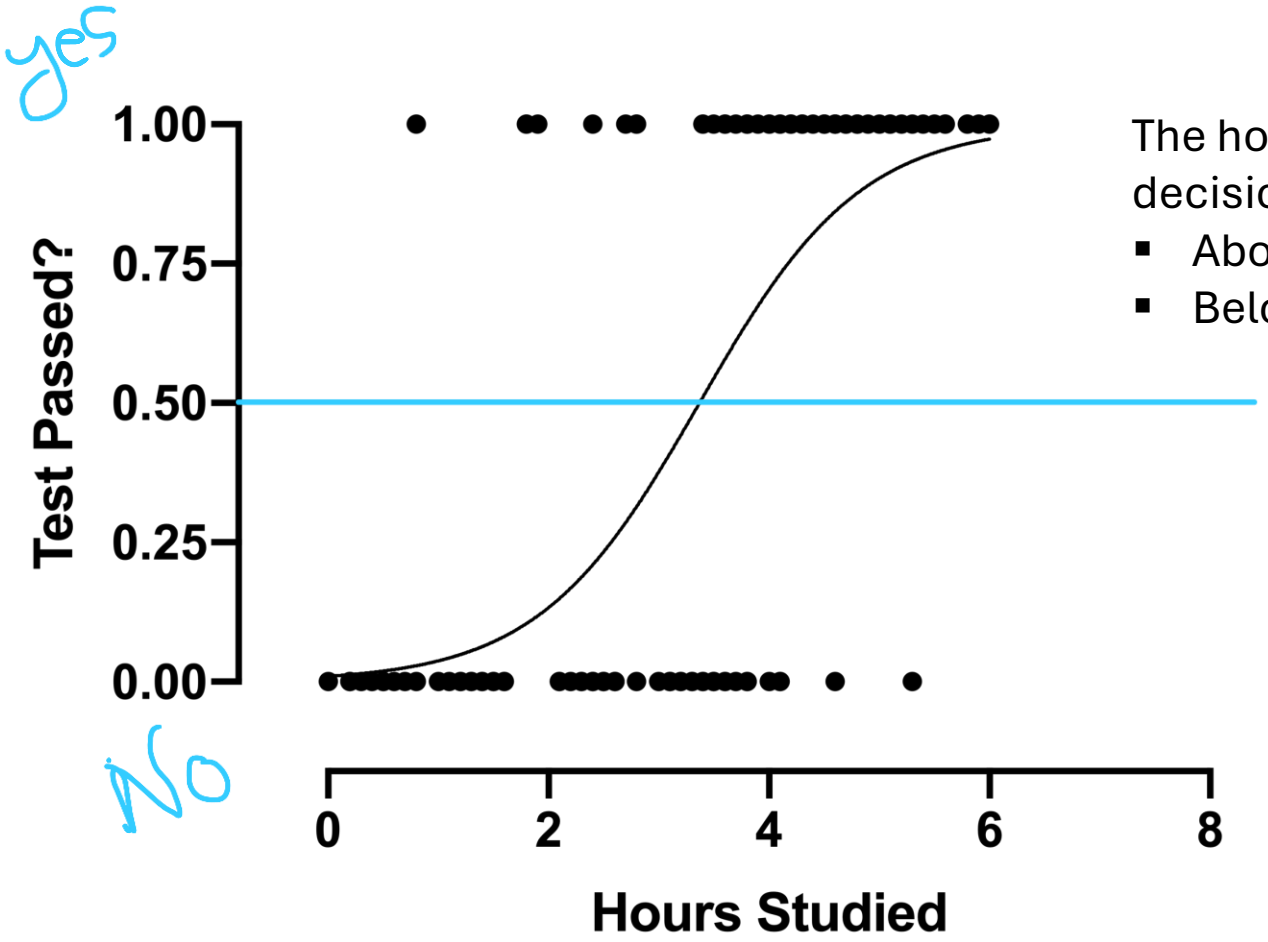
This part is **linear**, just like linear regression.

❌ **It doesn't stop there.**

Instead of outputting z directly (which could be any number from −∞ to +∞), logistic regression **plugs z into a sigmoid (logistic) function** to produce a value between 0 and 1:

$$\text{Output} = \frac{1}{1 + e^{-z}}$$

- Linear regression → direct numeric output (no squashing).
- Logistic regression → linear combination PLUS sigmoid to get a probability.



The horizontal **blue line** at 0.5 is the decision boundary:

- Above 0.5 → Predict "Yes" (pass).
- Below 0.5 → Predict "No" (fail).